

Position Sensorless for Controlling Brushless DC Motor Drives Based on Sliding Mode and RLS Estimators Using NSGA-II Algorithm Optimization

M. Shafiei, M. Bahrami Kouhshahi, M. B. B. Sharifian, M. R. Feyzi

Abstract – This paper presents a direct torque control (DTC) technique for Brushless DC (BLDC) motors with non-sinusoidal back electromotive force. DTC has some benefits such as faster torque response and reduced torque ripple for driving the BLDC motors. Because of ignoring stator flux linkage in DTC in constant torque region, BLDC motors driving is not complicated. The Sliding mode which is robust to parameter uncertainties can be used to estimate the back-EMF and generate the torque. It is employed to estimate the back-EMF waveform in a BLDC motor using only the measurements of the stator currents. The main advantages of the proposed estimator are: it is insensitive to switching noise and no filtering is required. Due to uncertainties of BLDC motor the moment of inertia and viscous damping coefficient are online identified simultaneously using recursive least square estimator. From the identified parameters the load torque is estimated. The estimated load torque is substituted into the mechanical dynamic equation to estimate the rotor speed. A modified version of Non-Dominated Sorting Genetic Algorithm (NSGA-II) is used as an effective optimization tool for regulating the PID controller parameters of speed controller and selection of four sliding mode observer gains. **Copyright** © 2011 Praise Worthy Prize S.r.l. - All rights reserved.

Keywords: Brushless DC (BLDC) Motor, Direct Torque Control (DTC), Recursive Least Square (RLS), Sliding Mode Observer, Non-Dominated Sorting Genetic Algorithm (NSGA-II)

Nomenclature

B	Viscous damping coefficient	L_s, M	Self and mutual inductances
e_a, e_b, e_c	Phase back-EMF	L	Phase inductance
$e_{sq}, e_{sd}, \hat{e}_{sq}, \hat{e}_{sd}$	Actual and estimated stationary reference frame back-EMFs	M_p, t_r, t_s	Overshoot, rise time and settling time of PID controller response
e_1, e_2, e_3, e_4	Error between actual and estimated stationary reference frame stator currents and back-EMFs	P	Number of poles
E_{ss}	Steady-state error of PID controller response	R	Phase resistance
F_{st}, T_{st}	Flux and torque error	$S(e)$	Sliding mode surface
$f_1, f_2, \dots, f_k$	k objective functions for multi-objective problem	$T_e, \hat{T}_e$	Actual and estimated electromagnetic torque
$g(v), h(v)$	Constraints for multiobjective problem	T_L	mechanical torque
i_a, i_b, i_c	Phase currents	TI, TD	Torque increase and torque decrease
i_{sq}, i_{sd}	Stationary reference frame stator currents	u_a, u_b, u_c	Phase voltages
J	Moment of inertia	u_1, u_2, y_1, y_2	Input and output variables
$k_{s1}, k_{s2}, k_{s3}, k_{s4}$	Sliding mode observer parameters	x_1, x_2, x_3, x_4	State variables
k_p, k_i, k_d	Proportional, integral and derivative gains of PID controller	α_1, α_2	system parameters
		β	Weight factor
		$\theta_e, \hat{\theta}_e$	Actual and estimated electrical rotor position
		λ	Forgetting factor
		v	Vector of control variables in multiobjective problem
		$\omega_e, \hat{\omega}_e$	Actual and estimated electrical speed of the rotor

ω_r	Mechanical and electrical speed of the rotor
$\succ$	Domination operator

I. Introduction

The Brushless DC (BLDC) motor is a type of electric motor, which is AC synchronous and looks quite similar to the DC motor. BLDC motors have come to dominate many applications such as home appliances, automobiles, information technology equipment, industries, transportation, aerospace, defense equipment, power tools, vision, sound equipment, and laboratory medical equipment in wide range of power. Due to better speed versus torque characteristics, high dynamic response, high efficiency, long operating life, noiseless operation and higher speed ranges, BLDC motors excel brushed DC and induction motors.

Direct Torque Control (DTC) method proposed in [1], [2] has been utilized to drive the BLDC motors. DTC, which is also known as conventional DTC, has the features like fast torque response, simple and robust design. These features have made it popular in industrial applications. This control method operates in two phase conduction mode which is simplified to just a torque controlled drive by intentionally keeping the stator flux linkage amplitude almost constant by eliminating the flux control in the constant torque region. Due to sharp changes in commutation region, the amplitude of stator flux linkage cannot easily be controlled.

For BLDC motor, only six discrete rotor positions per electrical cycle are needed to feed a rectangular current in phase with the back-EMF. For optimum performance, commutation has to be carried out at precise rotor positions. Hall sensors are normally used if good commutation performance is essential right down to zero speed. Hall sensors also allow smooth start-up [3]. However, it is a well known fact that these sensors have a great number of drawbacks; they increase the cost and need special mechanical arrangements to be mounted, especially in small motors. Further, Hall sensors are temperature-sensitive and hence limit the operation of the motor. They can reduce the system reliability because of their extra components and wiring. Consequently, there has been significant effort on practical implementation of sensorless techniques for determination of commutation instants.

These methods are based on a variety of strategies, including terminal voltage sensing, back-EMF integration, third harmonic of the back-EMF sensing, detection of freewheeling diode, flux calculation based methods and etc. [4]-[11].

Most popular and practical sensorless drive methods for BLDC motors rely on speed-dependent back-EMF. Since the back-EMF is zero or is undetectably small at standstill and low speeds, it is not possible to use the back-EMF sensing method in the low speed range. Also, the estimated commutation points that are shifted by 30

from zero crossing of back-EMFs have position error in transient state. To sense the third harmonic of the back-EMF, an external hardware circuit is required. This method also suffers from integration disadvantages at low speed. The most serious drawback of detection of freewheeling diode is the use of six isolated power supplies for the comparator circuit to detect the current flowing in each freewheeling diode. The flux estimation method also has significant estimation error at low speed, in which the voltage equation is integrated in a relatively large period of time [12]. To overcome the above drawbacks, some strategies for sensorless control of BLDC motor based on sliding mode observer and Extended Kalman Filter (EKF) are presented in [13]-[18]. EKF has some disadvantages such as, complicated structure, computationally tedious to implement and sensitive to the noise. Sliding mode observer is used for linear and nonlinear systems, and in many applications such as robotics, mobile robots, motors and converters. This type of observer is very useful and was developed for many reasons such as, working with reduced observation error dynamics, possibility of obtaining a step-by-step design, finite time convergence for all the observables states, designing an observer for non-smooth systems, and robustness under parameter variations is possible [19]. This paper proposes a sliding-mode observer to estimate the trapezoidal back-EMF of a BLDC motor.

In [20] recursive least square (RLS) is employed to identify the mechanical rotor inertia constant and the viscous damping coefficient. Both identified parameters are applied to estimate the load torque. By substituting the estimated load torque into mechanical dynamic equation the estimated rotor speed is achieved. The estimated back-EMF and rotor speed are in turn used to achieve estimated electromagnetic torque and rotor position using only the measurements of the stator currents. The proposed method overcomes some disadvantages that mentioned above.

To achieve the best response, the optimal PID controller parameters and sliding mode gains must be applied. This is a multiobjective problem (MOP). For this problem, maybe there isn't a best solution with respect to all objectives. A modified version of Non-Dominated Sorting Genetic Algorithm, NSGA-II, was developed to solve MOP problem [21]. Effectiveness of proposed method can be deduced from simulation results.

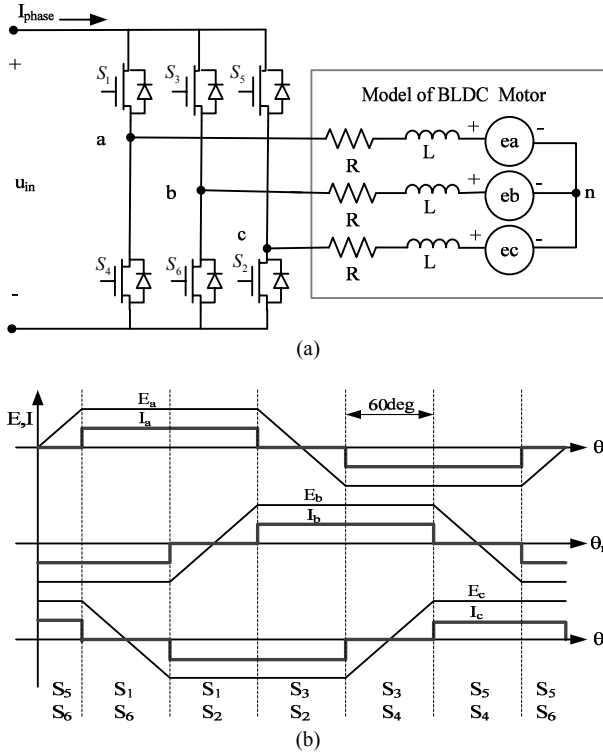
The system configuration, sensorless implementation, optimization algorithm and control method are presented in detail. Effectiveness of proposed method can be deduced from simulation results.

II. BLDC Motor Operation Principle

Brushed DC motors use mechanical commutator and brushes to achieve the commutation. However, BLDC motors are truly an inside-out DC Commutator with the

mechanical commutator replaced by an electronic switching converter.

The stator generates the magnetic field to make the rotor rotate. Hall Effect sensors detect the rotor position as the commutating signals. Therefore, the BLDC motors use permanent magnet instead of coil in the armature and do not need brushes. For the three-phase BLDC motors the back-EMF and phase current waveforms with 120° conduction mode are shown in Fig. 1, [5]-[7].



Figs. 1. (a) Configuration of BLDC motor drive system, (b) back EMF pattern and reference current generation

The analysis of a BLDC motor is represented as the following equations:

$$\begin{bmatrix} u_a \\ u_b \\ u_c \end{bmatrix} = \begin{bmatrix} R & 0 & 0 \\ 0 & R & 0 \\ 0 & 0 & R \end{bmatrix} \begin{bmatrix} i_a \\ i_b \\ i_c \end{bmatrix} + \begin{bmatrix} L_s - M & 0 & 0 \\ 0 & L_s - M & 0 \\ 0 & 0 & L_s - M \end{bmatrix} \frac{d}{dt} \begin{bmatrix} i_a \\ i_b \\ i_c \end{bmatrix} + \begin{bmatrix} e_a \\ e_b \\ e_c \end{bmatrix} \quad (1)$$

$(L = L_s - M)$

where u_a , u_b and u_c are the phase voltages, i_a , i_b and i_c are the phase currents, e_a , e_b and e_c are the phase back-EMFs, R is the phase resistance, L_s , M , and L are self, mutual, and phase inductances, respectively.

The electromagnetic torque can be obtained as:

$$T_e = \frac{1}{\omega_r} (e_a i_a + e_b i_b + e_c i_c) \quad (2)$$

where ω_r is the mechanical speed of the rotor.

The dynamic equation:

$$\frac{d}{dt} \omega_r = \frac{1}{J} (T_e - T_L - B \omega_r) \quad (3)$$

where B is the viscous damping coefficient, J is the moment of inertia of the drive and T_L is mechanical torque. The electrical speed ω_e related to the mechanical speed for a motor with P numbers of poles is:

$$\omega_e = \frac{P}{2} \omega_r \quad (4)$$

III. Direct Torque Control of BLDC Motor Drive using Two – Phase Conduction Mode

DTC was originally developed for induction machine drives, and controls the flux linkage and electromagnetic torque directly, considering the electrical machine, the power electronic inverter, and the control strategy at the system level. A relationship is established among the torque, the flux and the optimal inverter switching to achieve a fast torque response [1]. Switching losses are minimized because the transistors are switched only when it is needed to keep torque within hysteresis band.

The key issue in the DTC of a BLDC motor drive in the constant torque region is to estimate the electromagnetic torque correctly. For surface-mounted BLDC motors the back-EMF waveform is non-sinusoidal (trapezoidal), therefore equation (5) which is given in the stationary reference frame can be used for the electromagnetic torque calculation [1], [2], [8], [9]:

$$T_e = \frac{3}{2} \frac{P}{2} \frac{1}{\omega_e} (e_{sq} i_{sq} + e_{sd} i_{sd}) \quad (5)$$

where e_{sq} , e_{sd} and i_{sq} , i_{sd} are the stationary reference frame (dq-axes) motor back-EMFs and stator currents, respectively.

Fig. 2 shows the basic functional blocks used to implement the core of the DTC scheme. Four key blocks interact to provide the primary required control. These are the speed controller, the motor torque estimator, the torque comparator, and the switching logic which selects the inverter switching state. The torque comparator contain hysteresis control block to compare the torque reference with its corresponding actual value. The actual speed is compared to the speed command to form the torque command from the PI speed regulator.

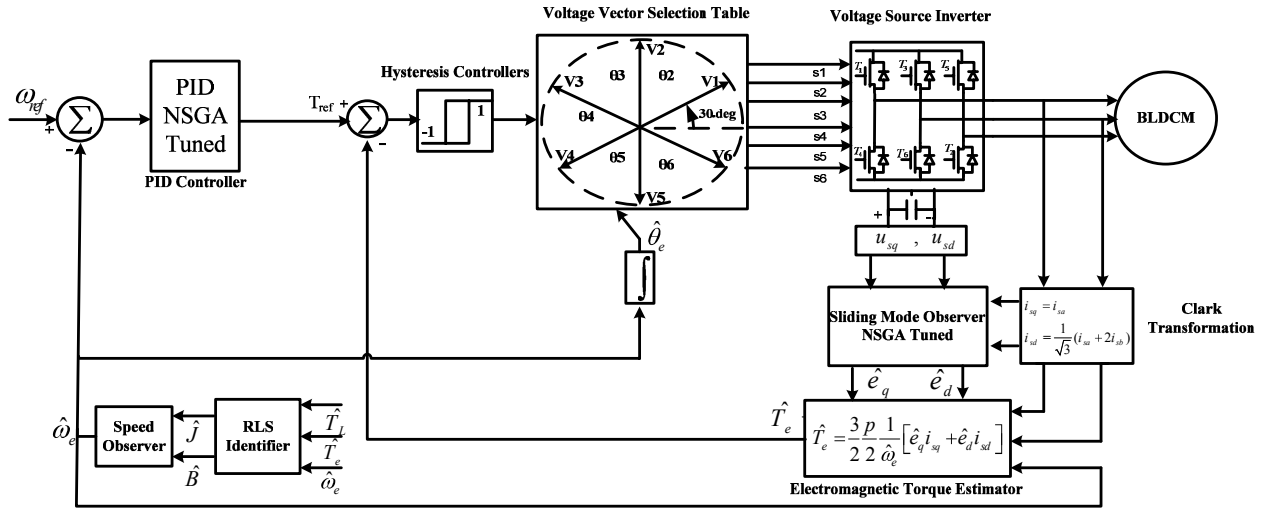


Fig. 2. Overall block diagram of the DTC of a BLDC motor drive

TABLE I
SWITCHING STATE SELECTOR FOR DTC, TORQUE INCREASE (TI), TORQUE DECREASE (TD) AND NO-CHANGE IN FLUX (F)

F_{st}	T_{st}	Sector					
		Θ_1	Θ_2	Θ_3	Θ_4	Θ_5	Θ_6
F	TI	$V_2(001001)$	$V_3(011000)$	$V_4(010010)$	$V_5(000110)$	$V_6(100100)$	$V_1(100001)$
	TD	$V_3(000110)$	$V_6(100100)$	$V_1(100001)$	$V_2(001001)$	$V_3(011000)$	$V_4(010010)$

Although in conventional DTC torque and flux are considered in the overall control system but three reasons are given for eliminating the flux control. First, since the magnitude of the phase back-EMF is below 50% of the dc-link voltage in the constant torque region, there is no need to control the flux amplitude. Second, with the two-phase conduction mode, sudden sharp dips in the stator flux linkage locus occur that complicate the control scheme. The size of these sharp dips is unpredictable.

Third, regardless of the stator flux linkage amplitude, the phase currents tend to match with the flat top portion of the corresponding trapezoidal back-EMF to generate constant torque.

As it can be seen from Table I, the switching pattern of the inverter can be determined according to the torque status from the output of hysteresis. Therefore, in the DTC of a BLDC motor drive the flux error F_{st} in the voltage vector selection look-up table is always selected as zero and only the torque error T_{st} is used depending on the error level of the actual torque from the reference torque. If the reference torque is bigger than the actual torque, within the hysteresis bandwidth, the torque error T_{st} is defined as "TI", otherwise it will be defined as "TD", as shown in Table I. By means of this method BLDC motor will be successfully driven.

IV. Observer Technique

IV.1. Back-EMF Estimation Based on Sliding Mode Observer

Sliding mode observers have unique properties, in that the ability to generate a sliding motion on the error

between the measured plant output and the output of the observer.

This ability ensures that a sliding mode observer produces a set of state estimates that are precisely commensurate with the actual output of the plant.

The BLDC motor equations are presented in (6):

$$\begin{aligned} \frac{di_{sq}}{dt} &= -\frac{R}{L}i_{sq} - \frac{1}{L}e_{sq} + \frac{1}{L}u_{sq} \\ \frac{di_{sd}}{dt} &= -\frac{R}{L}i_{sd} - \frac{1}{L}e_{sd} + \frac{1}{L}u_{sd} \\ \frac{de_{sq}}{dt} &= 0 \\ \frac{de_{sd}}{dt} &= 0 \end{aligned} \quad (6)$$

If the sampling period is significantly less than electrical and mechanical time constants then back-EMF value can be assumed to be constant during each sampling period so, $de_{sq}/dt = 0$, $de_{sd}/dt = 0$ [15].

System (6) can be written in the space as follows:

$$\begin{aligned} \dot{x}_1 &= -\alpha_1 x_1 + \alpha_2 (-x_3 + u_1) \\ \dot{x}_2 &= -\alpha_1 x_2 + \alpha_2 (-x_4 + u_2) \\ \dot{x}_3 &= 0 \\ \dot{x}_4 &= 0 \\ y_1 &= x_1 \\ y_2 &= x_2 \end{aligned} \quad (7)$$

where $x_1 = i_{sq}$, $x_2 = i_{sd}$, $x_3 = e_{sq}$ and $x_4 = e_{sd}$ are the

state variables, $u_1 = u_{sq}$ and $u_2 = u_{sd}$ are the input variables, y_1 and y_2 represent the output variables and $\alpha_1 = R/L$ and $\alpha_2 = 1/L$ are the system parameters.

The theory of sliding mode is represented in [22]. To decrease the chattering effects near the sliding mode surface, sign function is replaced with saturation function. Therefore, sliding mode observer is proposed as:

$$\begin{aligned}\dot{\hat{x}}_1 &= -\alpha_1 \hat{x}_1 + \alpha_2 (-\hat{x}_3 + u_1) + \\ &\quad + k_{s1} \text{sat}(x_1 - \hat{x}_1) \\ \dot{\hat{x}}_2 &= -\alpha_1 \hat{x}_2 + \alpha_2 (-\hat{x}_4 + u_2) + \\ &\quad + k_{s2} \text{sat}(x_2 - \hat{x}_2) \\ \dot{\hat{x}}_3 &= k_{s3} \text{sat}(x_1 - \hat{x}_1) \\ \dot{\hat{x}}_4 &= k_{s4} \text{sat}(x_2 - \hat{x}_2)\end{aligned}\quad (8)$$

By setting $e = x - \hat{x}$, error dynamic is represented in (9):

$$\begin{aligned}\dot{e}_1 &= -\alpha_2 e_3 + k_{s1} \text{sat}(e_1) \\ \dot{e}_2 &= -\alpha_2 e_4 + k_{s2} \text{sat}(e_2) \\ \dot{e}_3 &= k_{s3} \text{sat}(e_1) \\ \dot{e}_4 &= k_{s4} \text{sat}(e_2)\end{aligned}\quad (9)$$

Positive definite Lyapunov function is defined as sliding surface in (10):

$$S(e) = \frac{1}{2}(e_1^2 + e_2^2) \quad (10)$$

The time derivative of $S(e)$ is shown in (11):

$$\begin{aligned}\dot{S}(e) &= e_1 \dot{e}_1 + e_2 \dot{e}_2 \\ &= e_1 [-\alpha_2 e_3 + k_{s1} \text{sat}(e_1)] + \\ &\quad + e_2 [-\alpha_2 e_4 + k_{s2} \text{sat}(e_2)]\end{aligned}\quad (11)$$

The observer gains must be chosen so that the $S \cdot \dot{S} < 0$ limitation is satisfied. The NSGA-II is applied to tune observer gains. A block diagram of the sliding mode observer is given in Fig. 3.

IV.2. Viscous Damping Coefficient and Moment of Inertia Identification Based on RLS

Parameters B and J , in dynamic equation (3), are varying because of uncertainties, motor dynamics and different surrounding conditions. In this paper, RLS is used to estimate the parameters B and J . Equation (12) explains the motion control:

$$y(t) = \varphi^T(t) \cdot \theta(t) \quad (12)$$

where $y = (d/dt)\omega_r$, $\varphi(t) = \omega_r(T_e - T_L)$ and $\theta \left[\frac{B}{J} \frac{1}{J} \right]$.

Then, as presents in [20] and [23], the RLS rules are given by:

$$\hat{\theta}(t) = \hat{\theta}(t-1) + K(t)(y(t) - \varphi^T(t)\hat{\theta}(t-1)) \quad (13)$$

$$\begin{aligned}K(t) &= P(t-1)\varphi(t) \\ &= P(t-1)\varphi(t)(\lambda I + \varphi^T(t)P(t-1)\varphi(t))^{-1}\end{aligned}\quad (14)$$

$$P(t) = \frac{(I - K(t)\varphi^T(t))P(t-1)}{\lambda} \quad (15)$$

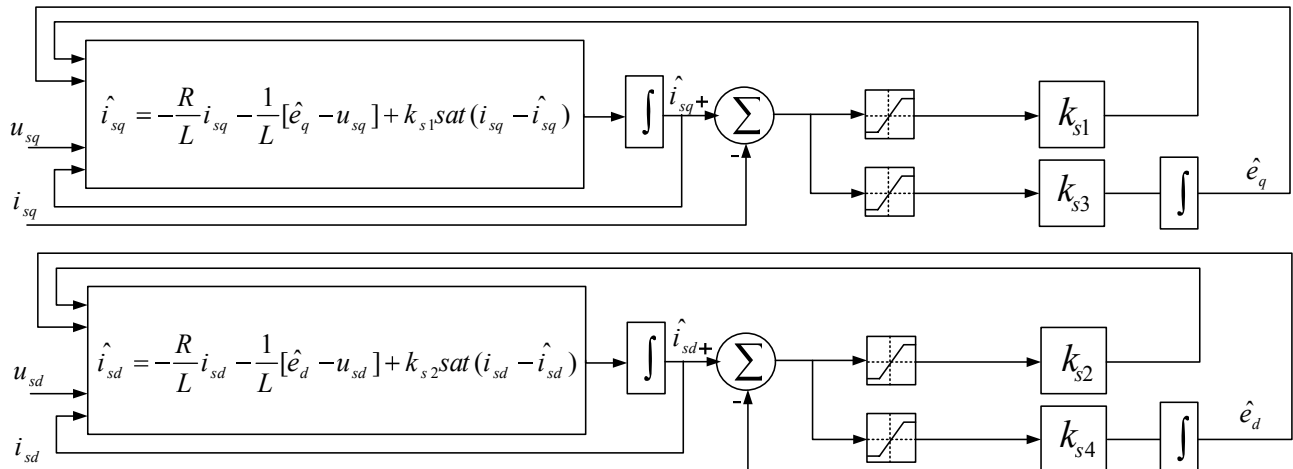


Fig. 3. Sliding mode observer block diagram

IV.3. Speed and Load Torque Estimation

As shown in Fig. 4 the estimated load torque can be achieved using $\hat{B}$ and $\hat{J}$. By substituting the estimated load torque in (3) the estimated rotor speed obtained. The estimated torque is also fed forward through a weighting factor, K , to realize a robust system.

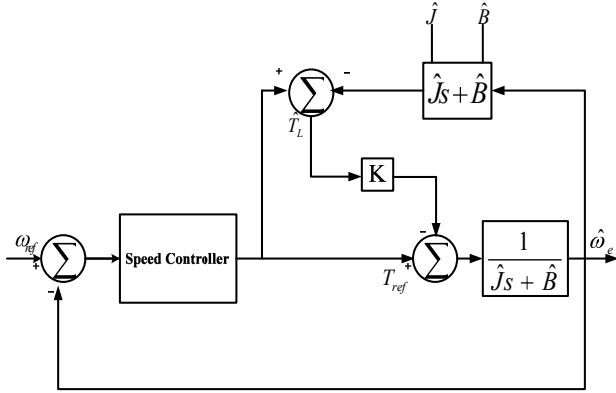


Fig. 4. Speed observer block diagram

Consider the back-EMFs and rotor speed that are estimated, the torque and electrical rotor speed can be calculated.

According to estimated back-EMF, estimated rotor speed and measured stator currents, the estimated torque is calculated as follow:

$$\hat{T}_e = \frac{3}{2} \frac{P}{2} \frac{1}{\hat{\omega}_e} (\hat{e}_q i_{sq} + \hat{e}_d i_{sd}) \quad (16)$$

Estimated electrical position of the rotor is calculated as follow:

$$\hat{\omega}_e = \frac{d}{dt} \hat{\theta}_e \quad (17)$$

V. Multiobjective Optimization Problem

Multiobjective optimization is an appropriate tool for handling different incommensurable objectives with conflicting/supporting relations or not having any mathematical relation with each other.

V.1. Basic Concepts

A multiobjective optimization problem could be written in the form:

$$\text{minimize } f(v) = \{f_1(v), f_2(v), \dots, f_k(v)\} \quad (18)$$

In the presence of constraints as:

$$\begin{aligned} g(v) &= \{g_1(v), g_2(v), \dots, g_k(v)\} \leq 0 \\ h(v) &= \{h_1(v), h_2(v), \dots, h_k(v)\} = 0 \end{aligned} \quad (19)$$

v is the vector of control variables. The solution of (18) is usually not unique. The concept of Pareto optimality may be explained in terms of a dominance relation. For a multiobjective problem having k objective functions to be simultaneously minimized, a solution v_1 is said to dominate the other solution v_2 if v_1 is better than v_2 for at least one objective if f_i and is not worse for any other f_j , where $j=1,2,\dots,k$ and $j \neq i$ as (20):

$$f_i(v_1) < f_i(v_2) \text{ and } f_j(v_1) \leq f_j(v_2) \Rightarrow v_1 \succ v_2 \quad (20)$$

Here, the symbol $\succ$ denotes domination operator. The above concept is used to find a set of non-dominated solutions in search space. The solutions that are non-dominated regarding the entire search space are called Pareto optimal solutions [24].

V.2. Objective Functions and Constraints

This problem has two objective functions which must be simultaneously minimized. These objective functions are defined for tuning PID controller parameters and sliding mode observer gains.

The transfer function of a PID controller is described as:

$$G(s) = k_p + \frac{k_i}{s} + k_d s \quad (21)$$

where k_p , k_i and k_d are the proportional, integral and derivative gains, respectively.

The system response can be evaluated by the overshoot (M_p), rise time (t_r), settling time (t_s), and steady-state error (E_{ss}), [25]. To improve the system performance by considering these parameters, the first objective function is defined as:

$$f_1(k_p, k_i, k_d) = (1 - e^{-\beta})(M_p + E_{ss}) + e^{-\beta}(t_s - t_r) \quad (22)$$

where β is the weight factor.

Decreasing rise time and settling time can be achieved by weight factor smaller than 0.7 and enhancing this parameter upper than 0.7, lead to reduction in overshoot and steady state error. The lower and upper limits of control variables are as follows:

$$\begin{aligned} 0 &< k_p < 100 \\ 0 &< k_i < 10 \\ 0 &< k_d < 10 \end{aligned} \quad (23)$$

According to (10) the second objective function can be defined as:

$$f_2(k_{s1}, k_{s2}, k_{s3}, k_{s4}) = \frac{1}{2}(e_1^2 + e_2^2) \quad (24)$$

As regards limitation $S \cdot \dot{S} < 0$ must be satisfied and according to (10) and (11), some constraints are extracted as follows:

- if $e_1 > 0$, the switching gain, $k_{s1} > -\alpha_2 e_3$
- if $e_1 < 0$, the switching gain, $k_{s1} > \alpha_2 e_3$
- if $e_2 > 0$, the switching gain, $k_{s2} > -\alpha_2 e_4$
- if $e_2 < 0$, the switching gain, $k_{s2} > \alpha_2 e_4$

In addition, two other constraints must be considered for k_{s3} and k_{s4} . According to concept of sliding mode, the system's behavior is on the sliding mode surface if $e_1 = \dot{e}_1 = e_2 = \dot{e}_2 = 0$. By substituting this in (9), the relation of e_3 with $\dot{e}_3$ and e_4 with $\dot{e}_4$ are extracted as:

$$\begin{cases} \dot{e}_3 = \frac{k_{s3}}{k_{s1}} \cdot \alpha_2 \\ e_3 = \frac{k_{s3}}{k_{s1}} \cdot \alpha_2 \\ \dot{e}_4 = \frac{k_{s4}}{k_{s1}} \cdot \alpha_2 \\ e_4 = \frac{k_{s4}}{k_{s1}} \cdot \alpha_2 \end{cases}$$

For sliding on surface S , the signs of e_3 , $\dot{e}_3$ and e_4 , $\dot{e}_4$ must be opposite. Therefore the last constraints are:

$$\begin{cases} \frac{k_{s3}}{k_{s1}} < 0 \\ \frac{k_{s4}}{k_{s1}} < 0 \end{cases}$$

V.3. Solution Method

NSGA-II is one of the most efficient multiobjective evolutionary algorithms using elitist approach which has great advantages in comparison to other evolutionary algorithms.

The general trend of NSGA-II is developed to find a set of the best PID controller and Sliding Mode Observer parameters are as; initially, a random parent population is created, each solution is assigned as fitness (or rank) equal to its non-domination level (1 is the best level, 2 is the next-best level and so on). Thus the minimization of fitness is assumed. In addition to fitness value a parameter called crowding distance is calculated for each individual.

The crowding distance is a measure of how close an individual is to its neighbors. Large average crowding distance will result in better diversity in the population. Parents are selected from the population by using binary tournament selection based on the rank and crowding distance. An individual is selected in the rank if it is less than the other or if crowding distance is greater than the

other one. The selected population generates offsprings from crossover and mutation operators. The population with the current population and current offsprings is sorted again based on non-domination and only the best N individuals are selected, where N is the population size.

The selection is based on the rank and crowding distance in the last solution [21], [26].

Fig. 5 summarizes the basic steps of the NSGA-II applied to this problem. Table II shows the optimal sliding mode and PID controller parameters that are obtained by NSGA-II. Also parameters of NSGA-II are given in this table.

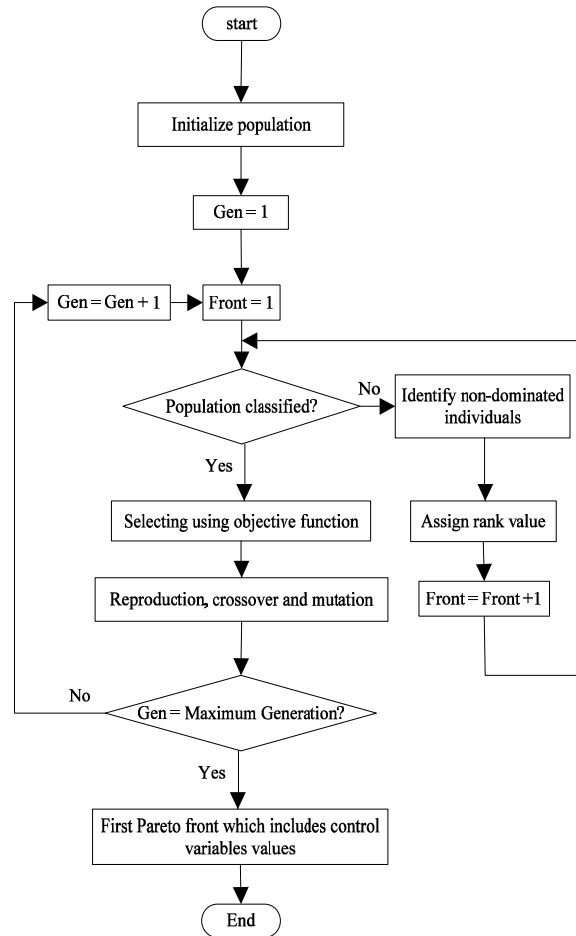


Fig. 5. Basic steps of NSGA-II

TABLE II
NSGA-II, OPTIMAL SLIDING MODE OBSERVER AND PID PARAMETERS

Parameter	Value
Total Number of Generations, Gen	50
Population Size, N	20
Proportional Gain, k_p	73.3254
Integrator Gain, k_i	0.51326
Derivative Gain, k_d	0.30096
k_{s1}	202.4226
k_{s2}	401.0967
k_{s3}	-25.2238
k_{s4}	-31.0089

VI. Simulation Results

In order to evaluate the effectiveness of proposed method, the drive system shown in Fig. 2 has been simulated by MATLAB/Simulink. Motor parameters used for the simulation are given in Table III.

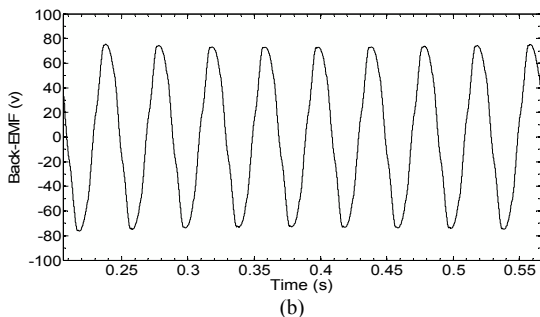
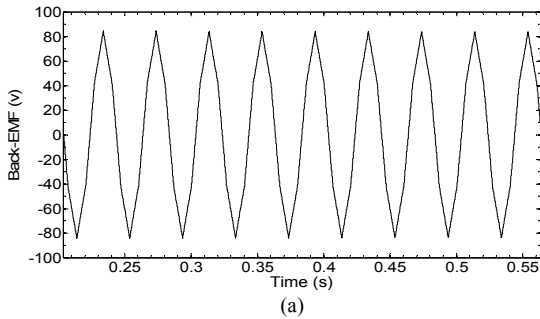
TABLE III
BLDC MOTOR PARAMETERS USED FOR SIMULATION

Parameter	Value
Number of Poles	2
DC Link Voltage	300 V
Rated Speed	1500 rpm
Phase Resistance	0.4 Ω
Phase Inductance	13 mH
Load Torque	3 N m
Moment of Inertia	0.004 kg m ²

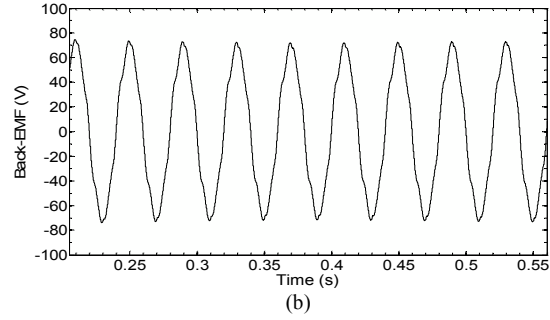
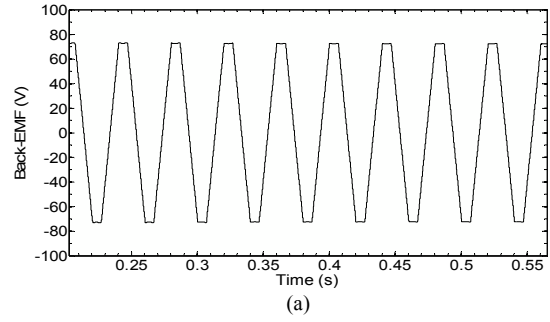
The actual and the estimated d-q frame back-EMFs are shown in Figs. 6 and 7. Figs. 6(a) and 7(a) are achieved from back-EMF shaping and Figs. 6(b) and 7(b) are estimated by sliding mode observer. It is obvious shown that the estimated back-EMFs are accurate and the sliding mode observer is a good option for back-EMFs estimation without requiring look up table and position sensors.

Estimated parameters $\hat{B}$ and $\hat{J}$ based on RLS algorithm are applied to the model, when the identification error becomes small enough. Figs. 8(a) and 8(b) represent the result of identified parameters $\hat{B}$ and $\hat{J}$.

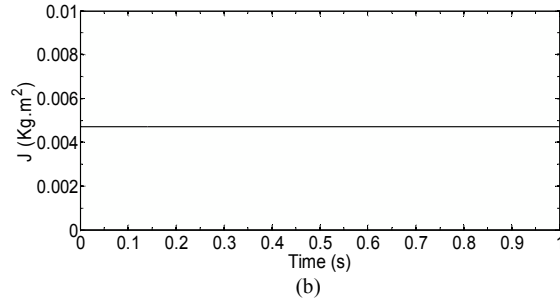
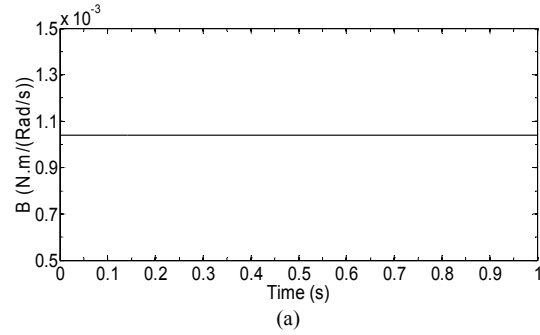
Fig. 9 shows the estimated load torque that is obtained according to section IV.3.



Figs. 6. Back-EMF (e_q), (a) Actual, (b) Estimated



Figs. 7. Back-EMF (e_d), (a) Actual, (b) Estimated



Figs. 8. Identified parameters, (a) Viscous damping coefficient, (b) moment of inertia

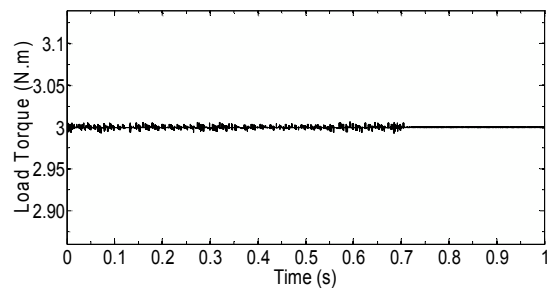
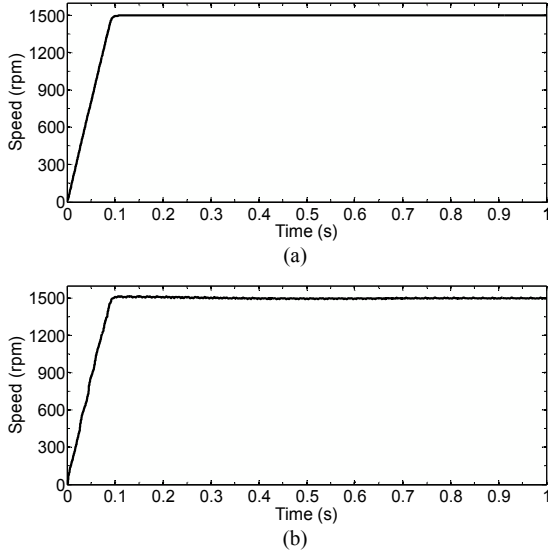


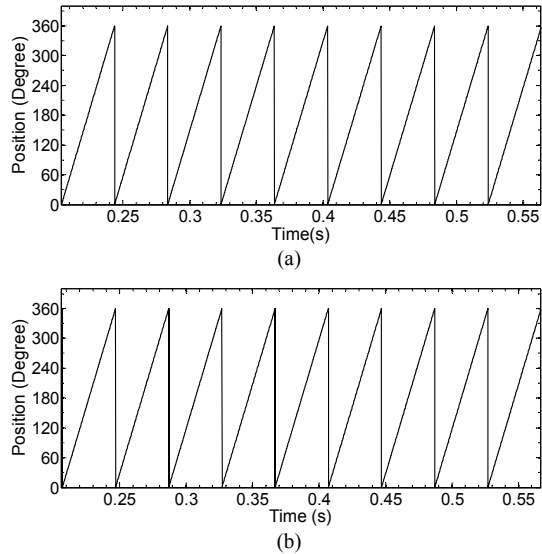
Fig. 9. Estimated load torque

Actual and estimated electrical rotor speeds are shown in Figs. 10. The trend calculation of estimated electrical rotor speed is given in subsection IV.3. By comparing Figs. 10(a) and 10(b), a good conformity of actual and estimated speed can be observed. Furthermore, speed rises steadily from stand still and reaches the steady state without any serious distortion.

The estimated electrical rotor position is obtained from electrical rotor speed. Figs. 11 show the actual and the estimated electrical rotor position.



Figs. 10. Rotor speed, (a) Actual, (b) Estimated



Figs. 11. Electrical rotor position, (a) Actual, (b) Estimated

Utilizing estimated rotor position instead of actual one can be ensured through good conformity of actual and estimated positions. As shown in Figs. 10 and 12, sectors of switching state selector can be determined using estimated rotor position without causing any tribulation in performance of DTC.

The stator phase currents are depicted in the Fig. 12. Despite elimination of position sensors, the waveforms of phase currents are appropriate. Fig. 13 shows estimated electromagnetic torque that is calculated by using estimated back-EMF, estimated rotor speed and measured stator current.

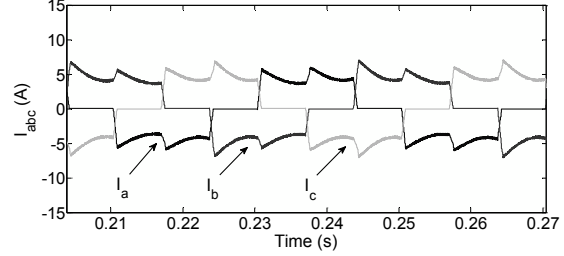


Fig. 12. Steady state motor phase currents

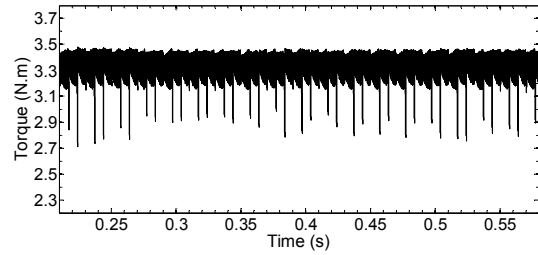


Fig. 13. Estimated electromagnetic torque

VII. Conclusion

The proposed position sensorless scheme first employed a RLS algorithm to identify the uncertain parameters of the plant. Then these parameters were used to estimate the rotor speed through a speed observer. Additionally, the sliding mode observer was applied to estimate the back-EMFs of BLDC motor. These estimated rotor speed and back-EMFs then were used to estimate the electromagnetic torque. Therefore, the proposed position sensorless scheme is unaffected by uncertainties, and since sliding mode back-EMF estimator is not speed-dependent, the proposed scheme has no problem to estimate the speed from stand still to steady state. DTC offers some advantages such as simple algorithm, simplicity to implement, faster torque response, reduced torque ripple and less sensitivity to the variations of parameters, so the proposed system has utilized DTC methods, in order to benefit from the mentioned advantages. The effectiveness of proposed sensorless method depends on proper selection of sliding mode observer and PID controller parameters; therefore NSGA-II algorithm was applied to tune these parameters, considering the relation between PID controller and the sliding mode observer.

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Authors' information



Mehdi Shafiei was born in Shiraz, Iran in 1987. He received the B.Sc degree in Control engineering from the Shiraz University of Technology, Shiraz, Iran in 2009 and M.S degree in electrical machine and drives, Tabriz, Iran in 2011.

His research interests include drive and design of permanent-magnet brushless DC machines, design of controller for power system stabilizer (PSS), intelligent algorithms, and adaptive and robust controller.

Mr. Shafiei is a professional member of the Institute of Electrical and Electronics Engineers (IEEE).

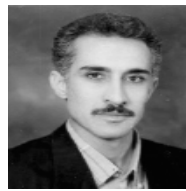


Mojtaba Bahrami Kouhshahi was born in Sirjan, Iran in 1984. He received the B.Sc. degree in Power engineering from the Shahid Bahonar University of Kerman, Kerman, Iran in 2008.

He is currently working on his M.Sc. degree at the University of Tabriz, Tabriz, Iran. His research interests include drive and design of electrical machines.

He currently focuses on drive and design of Permanent-Magnet Brushless DC machines.

Mr. Bahrami is a professional member of the Institute of Electrical and Electronics Engineers (IEEE).



Mohammad Bagher Bannae Sharifian (1965) Studied Electrical Power Engineering at the University of Tabriz, Tabriz, Iran. He received the B.Sc. and M.Sc. degrees in 1989 and 1992 respectively from University of Tabriz. In 1992 he joined the Electrical Engineering Department of the University of Tabriz as a lecturer. He received the Ph.D. degree in Electrical

Engineering from the same University in 2000. In 2000 he rejoined the Electrical Power Department of Faculty of Electrical and Computer Engineering of the same university as Assistant Professor. He is currently Professor of the mentioned Department. His research interests are in the areas of design, modeling and analysis of electrical machines, transformers, liner electric motors, and electric and hybrid electric vehicle drives.



Mohamad Reza Feyzi received his B.Sc. and M.Sc. in 1975 from university of Tabriz in Iran with honor degree. He worked in the same university during 1975 to 1993. He started his Ph.D. work at the University of Adelaide, Australia in 1993. Soon after his graduation, he rejoined to the University of Tabriz. Currently, he is a professor in the same university. His research interests are finite element analysis, design and simulation of electrical machines and transformers.